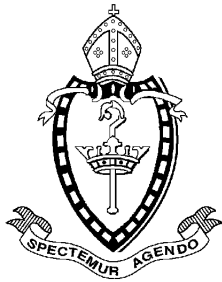


NEWCASTLE GRAMMAR SCHOOL

Student Number: _____



2011 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

Examination Date: Wednesday 17th August

Examiner: Mr. M. Brain

General Instructions

- Reading Time - 5 minutes
- Working Time - 3 hours
- Write using a blue or black pen
- Write your student number on every booklet
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Each question attempted is to be returned in a separate Writing Booklet clearly marked Questions 1 etc.
- If required, additional booklets may be requested

Total marks - 120

- Attempt Questions 1- 8
- All questions are of equal value

Question 1 (Start a new booklet)

Marks

a) Find

3

(i) $\int \frac{dx}{x^2 - 16x + 60}$

(ii) $\int \frac{dx}{x^2 - 16x + 80}$

b) Evaluate

8

(i) $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$

(ii) $\int_0^{\pi} e^x \cos x dx$

c) Use the substitution $u = x - 2$ to find $\int \frac{2x}{\sqrt{4x - x^2}} dx$

4

Question 2

(Start a new booklet)

Marks

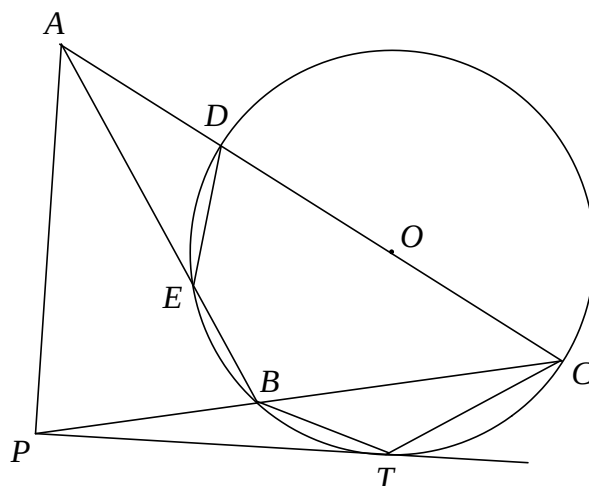
- a) (i) Express $\frac{-1+i}{\sqrt{3}+i}$ in mod-arg form **3**
- (ii) Hence express $\cos \frac{7\pi}{12}$ in surd form
- b) Evaluate $\arg((2+i)\overline{w})$ given that $w = -1-3i$ **1**
- c) (i) On an Argand diagram shade the region where both $|z-(1+i)| \leq 1$ and $0 \leq \arg(z-(1+i)) \leq \frac{\pi}{4}$ **4**
- (ii) Find the sets of values of $|z|$ and $\arg z$ for the points in the shaded region
- d) z_1 and z_2 are two complex numbers such that $\frac{z_1+z_2}{z_1-z_2} = 2i$ **7**
- (i) On an Argand diagram show vectors representing z_1, z_2, z_1+z_2 and z_1-z_2
- (ii) Show that $|z_1| = |z_2|$
- (iii) If α is the angle between the vectors representing z_1 and z_2 show that $\tan \frac{\alpha}{2} = \frac{1}{2}$
- (iv) Show that $z_2 = \frac{1}{5}(3+4i)z_1$

Question 3 (Start a new booklet)

Marks

- a) A is a point outside a circle with centre O . P is a second point outside the circle such that $PT=PA$ where PT is a tangent to the circle at T . AO cuts the circle at D and C . PC cuts the circle at B . AB cuts the circle at E .

6



Copy the diagram into your answer booklet

- (i) Show that $\triangle PBT$ is similar to $\triangle PTC$
- (ii) Show that $\triangle APB$ is similar to $\triangle CPA$
- (iii) Hence show that DE is parallel to AP
- b) (i) On the same number plane sketch the graphs of $y = |x| - 2$ and $y = 4 + 3x - x^2$
- (ii) Hence, or otherwise, solve $\frac{|x| - 2}{4 + 3x - x^2} > 0$

3

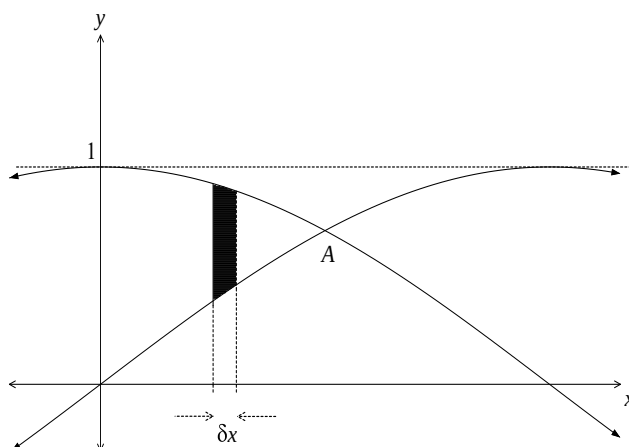
Question 3 continued on next page

Question 3 continued

Marks

- c) The area between $y = \sin x$ and $y = \cos x$, from the y -axis to the point of intersection, A , is rotated about the line $y = 1$

4



- (i) Find the co-ordinates of point A
- (ii) Calculate the generated volume of revolution
- d) Solve the equation $8x^4 + 44x^3 + 54x^2 + 25x + 4 = 0$ given that it has a triple root

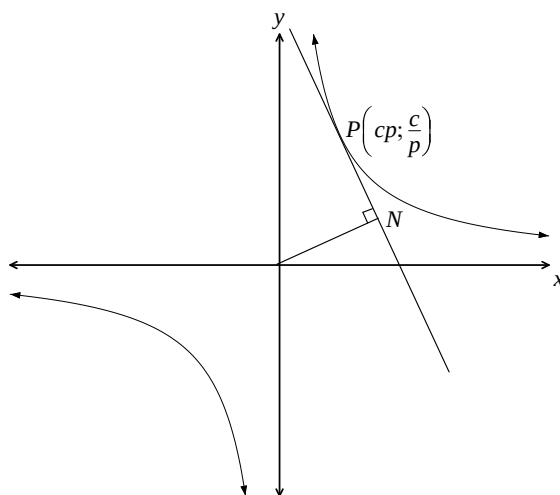
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Question 4

(Start a new booklet)

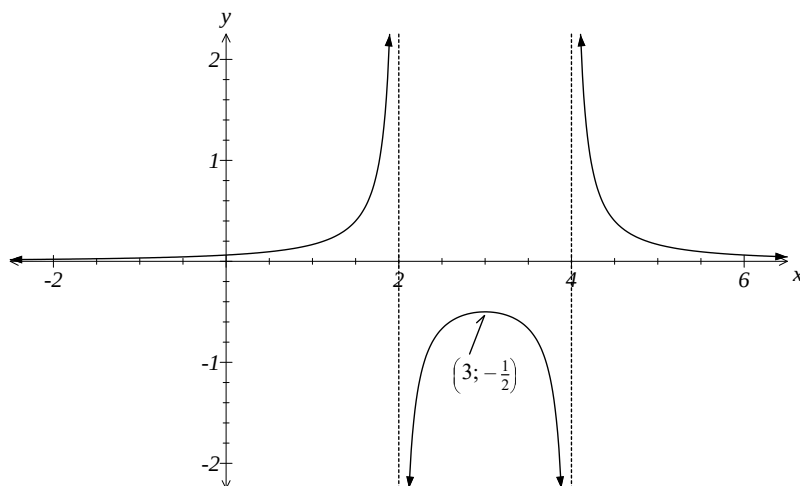
Marks

- a) Find all the roots of $P(x) = x^4 - 8x^3 + 39x^2 - 122x + 170$ given that $3 - i$ is one of the roots **4**
- b) The line through the origin which is perpendicular to the tangent at $P\left(cp, \frac{c}{p}\right)$ to the rectangular hyperbola $xy = c^2$ meets the tangent at N . **7**



Show that the locus of N has the equation $(x^2 + y^2)^2 = 4c^2xy$

- c) Give a possible equation for the graph below: **4**



Question 5 (Start a new booklet)

Marks

- a) The equation of a curve is $x^2y^2 - x^2 + y^2 = 0$

7

- (i) Show that the numerical value of y satisfies $|y| < 1$
- (ii) Find the equations of the asymptotes
- (iii) Show that $\frac{dy}{dx} = \frac{y^3}{x^3}$
- (iv) Sketch the curve

- b) Sketch the graph of each equation on a separate number plane:

8

- (i) (1) $y = \sqrt{x^2}$ (2) $y = (\sqrt{x})^2$
- (ii) (1) $y = \ln(e^x)$ (2) $y = e^{\ln x}$
- (iii) (1) $y = \sin(\sin^{-1} x)$ (2) $y = \sin^{-1}(\sin x)$

Question 6

(Start a new booklet)

Marks

- a) The inequality $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ is true for any real positive numbers a, b and c . Given that $a+b+c=1$ show:

5

(i) $\frac{1}{abc} \geq 27$

(ii) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$

(iii) $(1-a)(1-b)(1-c) \geq 8abc$

- b) (i) Show that the area, A , of a regular pentagon of side length P is given by

4

$$A = \frac{5}{2} P^2 \frac{\sin^2 54^\circ}{\sin 72^\circ}$$

- (ii) The area enclosed by $y = x^2$ and $y = 3$ is the base of a solid. Cross-sections of the solid, parallel to the x -axis, are regular pentagons with one side of the pentagon on the base of the solid. Calculate the volume of the solid, correct to one decimal place

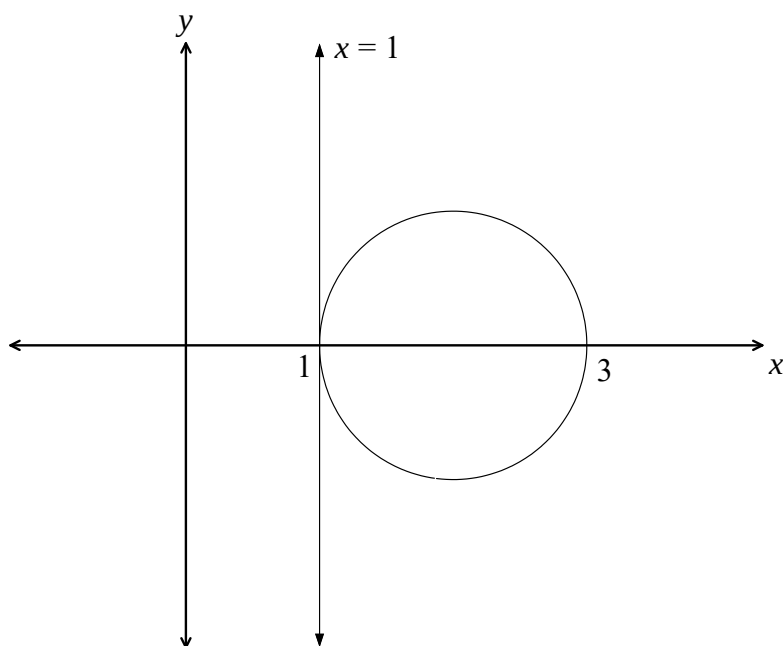
Question 6 continued on next page

Question 6 continued

Marks

- c) In the diagram below the circle with the equation $(x-2)^2 + y^2 = 1$ is drawn. The region bounded by the circle is rotated about the line $x = 1$

6



- (i) Use the method of cylindrical shells to show that the volume, V , of the solid so formed is given by

$$V = 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx$$

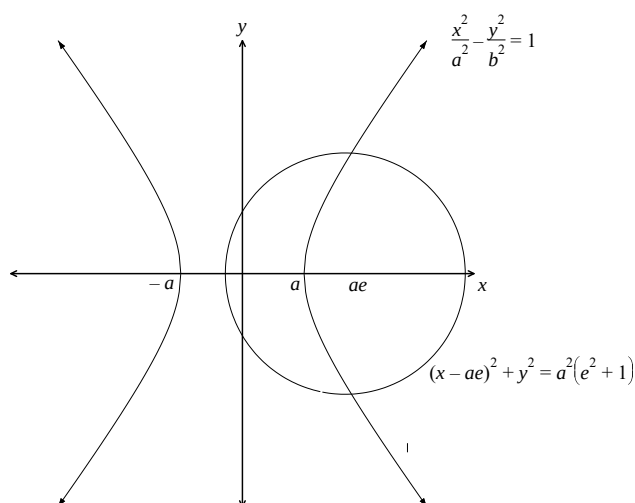
- (ii) By using the substitution $x-2 = \sin \theta$ calculate the volume of the solid formed

Question 7 (Start a new booklet)

Marks

a) For the diagram below:

8



- (i) Show that the tangent at $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has equation $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} - 1 = 0$
- (ii) Show that if the tangent at P is also a tangent to the circle with centre $(ae, 0)$ and radius $a\sqrt{e^2 + 1}$ then $\sec \theta = -e$
- (iii) Given that $\sec \theta = -e$, deduce that the points of contact, P and Q on the hyperbola, of the common tangents to the circle and the hyperbola are the extremities of a latus rectum ($x = -ae$) of the hyperbola and state the coordinates of P and Q
- (iv) Find the equations of the common tangents to the circle and the hyperbola and find the coordinates of their points of contact with the circle

b) (i) Show that $\frac{\sin(A+B) - \sin(A-B)}{2 \sin B} = \cos A$ 7

(ii) Hence show that

$$\cos x + \cos 3x + \cos 5x + \dots + \cos(2n-3)x + \cos(2n-1)x = \frac{\sin 2nx}{2 \sin x}$$

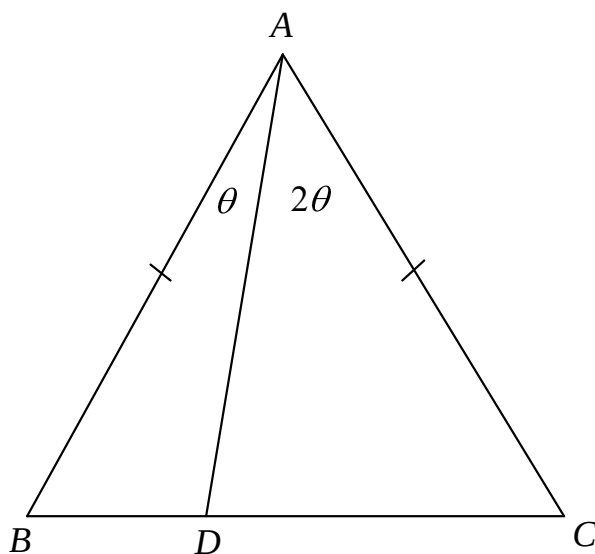
(iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin 8x}{\sin x} dx$

Question 8

(Start a new booklet)

Marks

- a) In the diagram below ABC is a triangle in which $AB = AC$ and $BC = 1$. D is the point on BC such that $\angle BAD = \theta$, $\angle CAD = 2\theta$ 6



(i) Letting $BD = x$ show that $\cos \theta = \frac{1-x}{2x}$

(ii) Hence show that $\frac{1}{3} < x < \frac{1}{2}$

- b) Let α, β and γ be the non-zero roots of $x^3 + 3px + q = 0$ 5

(i) Obtain the monic equation which has the roots $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}$ and $\frac{\beta\gamma}{\alpha}$

(ii) Show that if $\alpha\beta = \gamma$ then $(3p - q)^2 + q = 0$

c) (i) Prove that $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ **4**

(ii) Given that $y = 5 \sin(x + \alpha)$ prove, by mathematical induction,
that $\frac{d^n y}{dx^n} = 5 \sin\left(x + \alpha + \frac{n\pi}{2}\right)$ for $n \geq 1$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

① a) (i) $\int \frac{dx}{x^2 - 16x + 60}$

$$x^2 - 16x + 60 = (x-6)(x-10)$$

$$\therefore \int = \int \frac{dx}{(x-6)(x-10)}$$

Letting: $\frac{1}{(x-6)(x-10)} = \frac{A}{x-6} + \frac{B}{x-10}$

$$\therefore 1 = A(x-10) + B(x-6)$$

For $x = 10$: $1 = 4B \therefore B = \frac{1}{4}$

$x = 6$: $1 = -4A \therefore A = -\frac{1}{4}$

$$\therefore \int = \frac{1}{4} \int \left(\frac{1}{x-10} - \frac{1}{x-6} \right) dx$$

$$= \frac{1}{4} \{ \ln |x-10| - \ln |x-6| + C \}$$

$$\text{or} = \frac{1}{4} \left\{ \ln \left| \frac{x-10}{x-6} \right| + C \right\}$$

$$\text{or} = \ln \left| \frac{x-10}{x-6} \right|^{\frac{1}{4}} + C$$

②

(ii) $\int \frac{dx}{x^2 - 16x + 80}$

$$= \int \frac{dx}{(x-8)^2 + 16}$$

$$= \frac{1}{4} \tan^{-1} \left(\frac{x-8}{4} \right) + C$$

①

b) (i) Letting $\tan \frac{\theta}{2} = t$

$$\therefore \frac{dt}{d\theta} = \frac{1}{2} (\sec^2 \frac{\theta}{2})$$

$$= \frac{1}{2} (\tan^2 \frac{\theta}{2} + 1)$$

$$= \frac{1}{2} (t^2 + 1)$$

$$\therefore d\theta = \frac{2}{t^2 + 1} dt$$

$$\text{and } \sin \theta = \frac{2t}{t^2 + 1}$$

and at $\theta = \frac{\pi}{2}$: $t = \tan \frac{\pi}{4} = 1$

$\theta = 0$: $t = \tan 0 = 0$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$$

$$= \int_0^1 \frac{1}{1 + \frac{2t}{t^2 + 1}} \times \frac{2}{t^2 + 1} dt$$

$$= \int_0^1 \frac{2}{t^2 + 1 + 2t} dt$$

$$= \int_0^1 \frac{2}{(t+1)^2} dt$$

$$= 2 \int_0^1 (t+1)^{-2} dt$$

$$= 2 \left[\frac{-1}{t+1} \right]_0^1$$

$$= -2 \left(\frac{1}{1+1} - \frac{1}{1+0} \right)$$

$$= -2 \left(\frac{1}{2} - 1 \right)$$

$$= 1$$

④

(ii) $\int_0^{\pi} e^x \cos x dx = I$

$$u = e^x \quad v' = \cos x$$

$$u' = e^x \quad v = \sin x$$

$$= uv - \int v u'$$

$$= [e^x \sin x]_0^{\pi} - \int_0^{\pi} e^x \sin x dx$$

$$u = e^x \quad v' = \sin x$$

$$u' = e^x \quad v = -\cos x$$

$$= [e^x \sin x]_0^{\pi} - [-e^x \cos x]_0^{\pi} + \int_0^{\pi} e^x \cos x dx$$

$$\therefore I = [e^x \sin x + e^x \cos x]_0^{\pi} - I$$

$$\therefore 2I = [e^x (\sin x + \cos x)]_0^{\pi}$$

$$\therefore I = \frac{1}{2} \{ (e^{\pi} (0-1)) - (e^0 (0+1)) \}$$

$$= \frac{1}{2} (-e^{\pi} - 1)$$

④

c) $\int \frac{2x}{\sqrt{4x-x^2}} dx = \int \frac{2(u+2)}{\sqrt{4(u+2)-(u+2)^2}} du \quad \rightarrow u = x-2 \quad \therefore du = dx$

$$= 2 \int \frac{u+2}{\sqrt{4-u^2}} du$$

$$= 2 \int \frac{u}{\sqrt{4-u^2}} du + 4 \int \frac{1}{\sqrt{4-u^2}} du \quad \rightarrow u^2 = w$$

$$= 2 \left\{ \frac{1}{2} \int \frac{2u}{\sqrt{4-w}} dw \right\} + 4 \sin^{-1} \left(\frac{u}{2} \right) + C \quad \leftarrow dw = 2u du$$

$$= \int (4-w)^{-\frac{1}{2}} dw + 4 \sin^{-1} \left(\frac{u}{2} \right) + C$$

$$\begin{aligned}
 &= \frac{(4-u)^{\frac{1}{2}}}{-\frac{1}{2}} + 4 \sin^{-1}\left(\frac{u}{2}\right) + C \\
 &= -2 \sqrt{4-u^2} + 4 \sin^{-1}\left(\frac{u}{2}\right) + C \\
 &= -2 \sqrt{4-(x-2)^2} + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C \\
 &= -2 \sqrt{4x-x^2} + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C \quad (1)
 \end{aligned}$$

$$(2) \text{ a) (i) } \frac{-1+i}{\sqrt{3}+i} = \frac{\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}{2 \operatorname{cis} \frac{\pi}{6}}$$

$$= \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{3\pi}{4} - \frac{\pi}{6} \right)$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis} \left(\frac{7\pi}{12} \right) \quad (1)$$

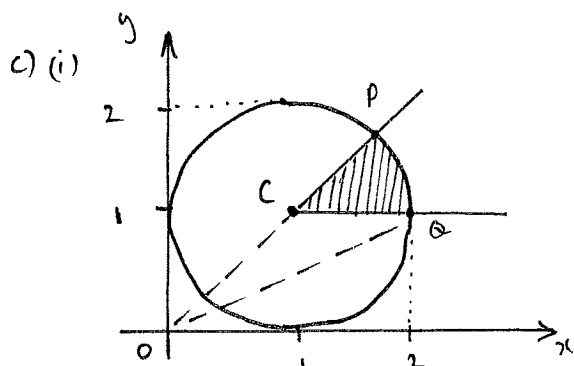
$$(ii) \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{1-\sqrt{3} + (1+\sqrt{3})i}{4}$$

$\therefore$ equating real parts of (i) and (ii):

$$\frac{1}{\sqrt{2}} \cos \left(\frac{7\pi}{12} \right) = \frac{1-\sqrt{3}}{4}$$

$$\therefore \cos \left(\frac{7\pi}{12} \right) = \frac{\sqrt{2}-\sqrt{6}}{4} \quad (2)$$

$$\begin{aligned}
 \text{b) } \arg((2+i)-\bar{w}) &= \arg((2+i)(-1+3i)) \\
 &= \arg(-5+5i) \\
 &= \frac{3\pi}{4} \quad (1)
 \end{aligned}$$



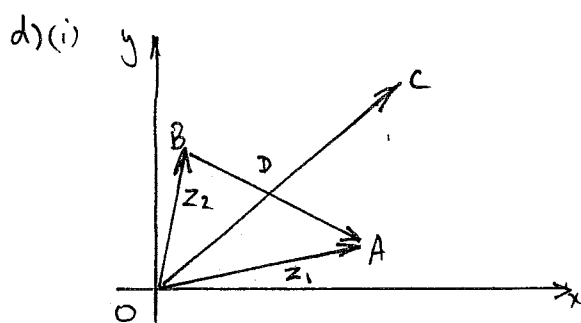
$$(ii) \quad OC < |z| < OP$$

$$\therefore \sqrt{2} < |z| < \sqrt{2}+1$$

$\arg z$ min at Q, $\arg z$ max at P

$$\therefore \tan^{-1} \frac{1}{2} < \arg z < \tan^{-1} 1$$

$$\text{or } \tan^{-1} \frac{1}{2} < \arg z < \frac{\pi}{4} \quad (2)$$



$$OA = z_1, OB = z_2, OC = z_1 + z_2, BA = z_1 - z_2$$

NOTE: AOB C is parallelogram
(by addition of vectors)

$$(ii) \arg \left(\frac{z_1 + z_2}{z_1 - z_2} \right) = \arg(2i)$$

$$\therefore \arg(z_1 + z_2) - \arg(z_1 - z_2) = \frac{\pi}{2}$$

$\therefore$ angle between diagonals of parallelogram = 90°

$\therefore$ AOB C is rhombus

$$\therefore |z_1| = |z_2| \quad (\text{QED}) \quad (2)$$

$$(iii) \left| \frac{z_1 + z_2}{z_1 - z_2} \right| = |2i|$$

$$\therefore \frac{|z_1 + z_2|}{|z_1 - z_2|} = 2$$

$$\therefore |OC| = 2 \times |AB|$$

$$\therefore OD = 2 \times AD \quad (1)$$

$$\text{and } \tan \frac{\alpha}{2} = \frac{AD}{OD} = \frac{AD}{2AD} = \frac{1}{2} \quad (\text{QED})$$

$$(iv) \frac{1}{5}(3+4i) = \frac{1}{5} \times 5 \operatorname{cis}(\tan^{-1} \frac{4}{3})$$

$$= \operatorname{cis}(\tan^{-1}(\frac{4}{3}))$$

$$\therefore |\frac{1}{5}(3+4i)| = 1$$

$$\arg(\frac{1}{5}(3+4i)) = \tan^{-1}(\frac{4}{3})$$

$\therefore z_1 \times (\frac{1}{5}(3+4i))$ is same modulus as z_1 , rotated through $\tan^{-1}(\frac{4}{3})$

$$\text{From diagram: } \tan \frac{\alpha}{2} = \frac{1}{2}$$

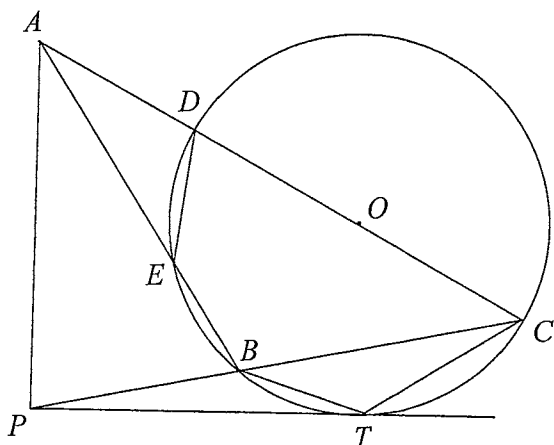
$$\therefore \tan \alpha = \frac{2 \times \frac{1}{2}}{1 - (\frac{1}{2})^2}$$

$$= \frac{4}{3}$$

$$\therefore \alpha = \tan^{-1}(\frac{4}{3}) \text{ and } |z_1| = |z_2|$$

$$\therefore z_2 = \frac{1}{5}(3+4i)z_1 \quad (\text{QED}) \quad (2)$$

(3)
a)



(i) $\angle BPT = \angle CPT$ (common) (A)
 $\angle PTB = \angle PCT$ ($\angle$ bet. tang. + chord) (A)
 $\therefore \angle PBT = \angle PTC$ ($\angle$ sum Δ) (A)

$$\therefore \Delta PBT \parallel \Delta PTC \quad (AAA)$$

(QED) (1)

(ii) $\angle APB = \angle CPA$ (common) (A)

$$\text{and } \frac{PB}{PT} = \frac{PT}{PC} \quad (\text{corres. sides } \parallel \Delta\text{'s, from (i)})$$

$$\text{but } PT = PA \quad (\text{given})$$

$$\therefore \frac{PB}{PA} = \frac{PA}{PC} \quad (s, s)$$

$$\therefore \Delta APB \parallel \Delta CPA \quad (S.A.S) \quad (\text{QED}) \quad (2)$$

(iii) $\angle PAE = \angle BCD$ (corres. $\angle$'s in $\parallel \Delta$'s, from (ii))

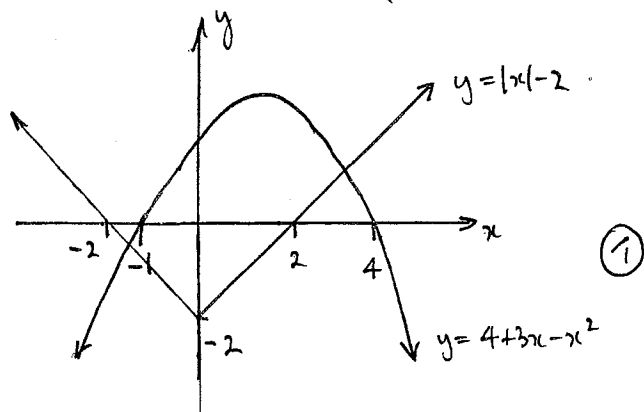
and $\angle BCD = \angle DEA$ (ext. $\angle$ of cyclic quad)

$$\therefore \angle PAE = \angle DEA$$

$$\therefore DE \parallel AP \quad (\text{alt. } \angle\text{'s are } =) \quad (\text{QED}) \quad (2)$$

$$b) (i) \quad y = |x| - 2 \quad y = 4 + 3x - x^2$$

$$= -(x+1)(x-4)$$



$$(ii) \text{ For } \frac{|x|-2}{4+3x-x^2} > 0$$

$$\text{i.e. } \frac{|x|-2}{4+3x-x^2} \text{ POSITIVE}$$

i.e. when BOTH graphs on same side of x -axis

$\therefore$ Solution is:

$$-2 < x < -1 \text{ or } 2 < x < 4. \quad (2)$$

c) (i) For A: $\sin x = \cos x$

$\therefore \tan x = 1$

$\therefore x = \pi/4, y = \frac{1}{\sqrt{2}}$

(ii) $A(x) = \text{area of annulus}$

$= \pi \{ (1 - \sin x)^2 - (1 - \cos x)^2 \}$

$\therefore V = \lim_{\delta x \rightarrow 0} \sum_a^b A(x) \delta x$

$= \pi \int_0^{\pi/4} (1 - 2\sin x + \sin^2 x - 1 + 2\cos x - \cos^2 x) dx$

$= \pi \int_0^{\pi/4} (2\cos x - 2\sin x - \cos 2x) dx$

$= \pi [2\sin x + 2\cos x - \frac{1}{2}\sin 2x]_0^{\pi/4}$

$= \pi ((2 \times \frac{1}{\sqrt{2}} + 2 \times \frac{1}{\sqrt{2}} - \frac{1}{2}) - (0 + 2 - 0))$

$\therefore V = \pi (\frac{4}{\sqrt{2}} - \frac{1}{2})$
or $\pi (2\sqrt{2} - 0.5) \text{ u}^3$

d) For $P(x) = 8x^4 + 44x^3 + 54x^2 + 25x + 4 = 0$

$\therefore P'(x) = 32x^3 + 132x^2 + 108x + 25$

$\therefore P''(x) = 96x^2 + 264x + 108$

$= 12(4x+9)(2x+1)$

$\therefore (4x+9) \text{ or } (2x+1) \text{ is multiple factor}$

Now: $P(-\frac{1}{2}) = 8(-\frac{1}{2})^4 + 44(-\frac{1}{2})^3 + 54(-\frac{1}{2})^2 + 25(-\frac{1}{2}) + 4$
 $= 0$

$\therefore P(x) = (2x+1)^3 \times Q(x)$ by inspection
ie. $P(x) = (2x+1)^3 (x+4)$

$\therefore x = -\frac{1}{2} \text{ or } -4$

④ a) $P(x)$ has real coefficients

$\therefore$ roots in conjugate pairs

$\therefore 3-i \text{ is root} \Rightarrow 3+i \text{ is root}$

$\therefore (x-(3-i))(x-(3+i))$ is factor of $P(x)$

ie. $x^2 - 6x + 10$ is factor

$\therefore$ by division or by inspection:

$P(x) = (x^2 - 6x + 10)(x^2 - 2x + 17)$

For $x^2 - 2x + 17 = 0 : x = \frac{2 \pm \sqrt{-64}}{2}$
 $= 1 \pm 4i$

$\therefore$ Roots are: $3 \pm i, 1 \pm 4i$

b) $y = \frac{c^2}{x}$
 $\therefore y' = -\frac{c^2}{x^2}$

$\therefore$ Equation of tangent at P:

$y - y_p = -\frac{1}{p^2} (x - c_p)$

$\therefore x + p^2 y = 2cp \dots \dots (1)$

and: Equation of ON is

$y = p^2 x \dots \dots (2)$

$\therefore (2) \text{ into } (1) : x + p^2(p^2 x) = 2cp$

$\therefore x = \frac{2cp}{1+p^4}$

$\therefore y = \frac{2cp^3}{1+p^4}$

Now: $x(1+p^4) = 2cp$ and $p^2 = \frac{y}{x}$

$\therefore x^2 \left(1 + \frac{y^2}{x^2}\right)^2 = 4c^2 \left(\frac{y}{x}\right) \quad (x \text{ b.s. } x^2)$

$\therefore (x^2 + y^2)^2 = 4c^2 xy \quad (\text{Q.E.D.})$

c) Graph is of the form:

$$y = \frac{1}{a(x-2)(x-4)}$$

i.e. reciprocal graph of parabola

- with turning point at $x=3$

But at $x=3$:

$$-\frac{1}{2} = \frac{1}{a(3-2)(3-4)}$$

$$-\frac{1}{2} = -\frac{1}{a}$$

$$\therefore a = 2$$

$\therefore$ Equation is:

$$y = \frac{1}{2(x-2)(x-4)} \quad (4)$$

$$(5) a) (i) \quad x^2 y^2 - x^2 + y^2 = 0 \quad \dots\dots (1)$$

$$\therefore y^2(x^2+1) = x^2$$

$$\therefore y^2 = \frac{x^2}{x^2+1}$$

Now: $RHS \geq 0$ for all x

and Denominator > Numerator

$$\therefore RHS < 1$$

$$\therefore 0 \leq y^2 < 1$$

$$\text{i.e. } |y| < 1 \quad (QED) \quad 2$$

$$(ii) \text{ as } x \rightarrow \infty \quad y^2 \rightarrow 1$$

$$\therefore y \rightarrow \pm 1$$

$$\therefore \text{asymptotes: } y = \pm 1$$

(iii) By implicit differentiation:

$$x^2 2yy' + 2xy^2 - 2x + 2yy' = 0$$

$$\therefore y'(2x^2y + 2y) = 2x - 2xy^2$$

$$\therefore \frac{dy}{dx} = \frac{2x(1-y^2)}{2y(x^2+1)}$$

$$\text{but from (i)} \quad y^2(x^2+1) = x^2$$

$$\therefore y(x^2+1) = x^2/y$$

$$\text{and } x^2(1-y^2) = y^2$$

$$\therefore x(1-y^2) = y^2/x$$

$$\therefore \frac{dy}{dx} = y^2/x \div x^2/y$$

$$\text{i.e. } \frac{dy}{dx} = y^3/x^3$$

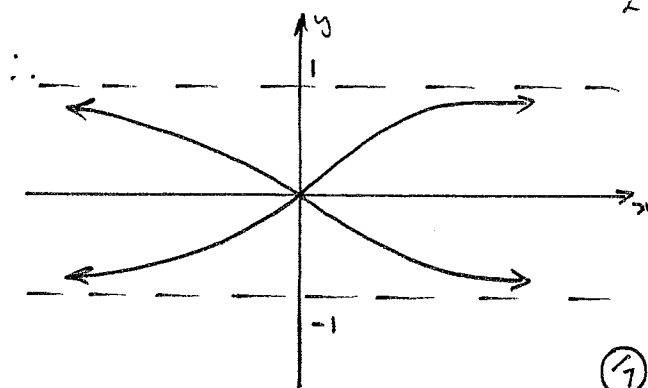
$$(iv) \text{ at } x=0 \quad y^2=0 \quad \therefore y=0$$

$$\therefore \text{ as } x \rightarrow 0, y \rightarrow 0$$

$$\text{i.e. } y \rightarrow x$$

$$\therefore \frac{dy}{dx} \rightarrow 1$$

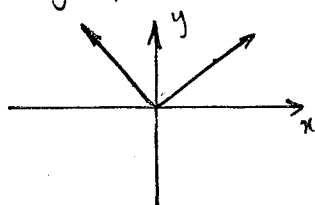
$$\text{as } x \rightarrow \pm \infty \quad \frac{dy}{dx} \rightarrow 0$$



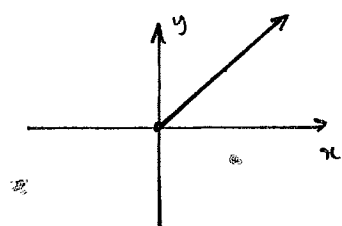
(NOTE: even in x and y

$\therefore$ symmetry about both axes)

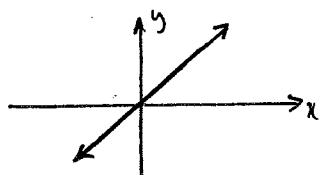
b) (i) (1) $y = \sqrt{x^2}$ } Domain: $x \in \mathbb{R}$
 $\therefore y = |x|$



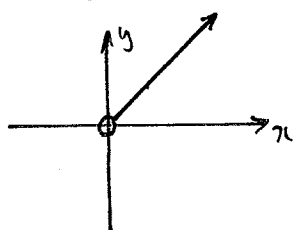
(2) $y = (\sqrt{x})^2$ } Domain: $x \geq 0$
 $\therefore y = x$



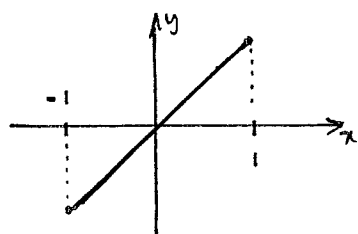
(ii) (1) $y = \ln(e^x)$ } Domain: $x \in \mathbb{R}$
 $\therefore y = x$



(2) $y = e^{\ln x}$ } Domain: $x > 0$
 $\therefore y = x$



(iii) (1) $y = \sin(\sin^{-1} x)$ } Domain: $-1 \leq x \leq 1$
 $\therefore y = x$



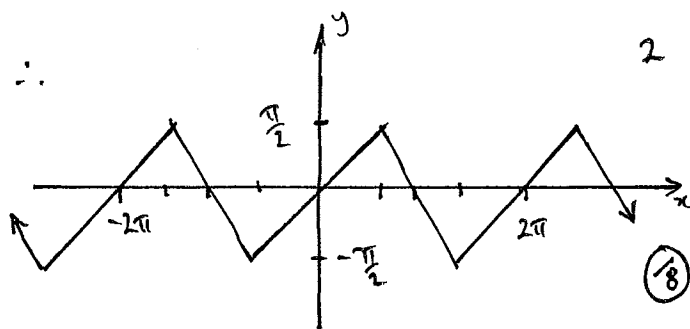
(2) $y = \sin^{-1}(\sin x)$ } Domain: $x \in \mathbb{R}$
 for $\sin x$
 } BUT: Domain: $-1 \leq \sin^{-1} x \leq 1$

$\therefore y = x$ } Domain: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

increasing graph: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$\therefore$ decreasing graph: $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$

AND periodic function.



⑥ a) (i) $\sqrt[3]{abc} \leq \frac{a+b+c}{3}$ } $a+b+c=1$
 $\therefore \leq \frac{1}{3}$

$\therefore abc \leq \frac{1}{27}$

$\therefore \frac{1}{abc} \geq 27$ (QED) ①

(ii) $\frac{1}{3} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq \sqrt[3]{\frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}}$

$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3 \sqrt[3]{\frac{1}{abc}}$

$\geq 3 \sqrt[3]{27}$

$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$ (QED) ②

a) (iii)

$$(1-a)(1-b)(1-c)$$

$$= 1 - (a+b+c) + (bc+ca+ab) - abc$$

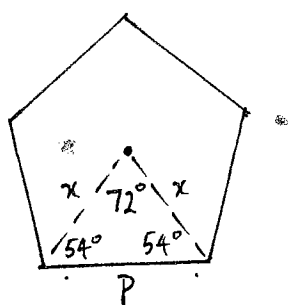
$$= (bc+ca+ab) - abc$$

$$= abc \left\{ \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) - 1 \right\}$$

$$\geq abc \{9 - 1\}$$

$$\therefore (1-a)(1-b)(1-c) \geq 8abc \quad (\text{QED}) \quad (2)$$

b) (i)



$$\text{Area of pentagon} = 5 \times \frac{1}{2} p^2 \sin 72^\circ$$

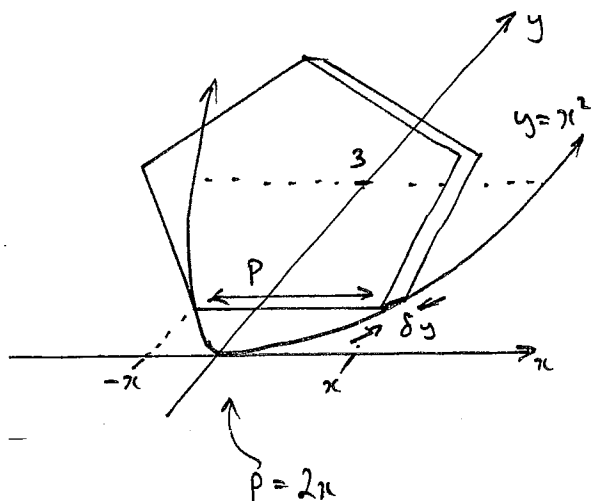
$$\text{and } \frac{x}{\sin 54^\circ} = \frac{p}{\sin 72^\circ}$$

$$\therefore \text{Area} = \frac{5}{2} \times \left(\frac{p \sin 54^\circ}{\sin 72^\circ} \right)^2 \times \sin 72^\circ$$

$$\therefore \text{Area} = \frac{5}{2} p^2 \frac{\sin^4 54^\circ}{\sin 72^\circ} \quad (\text{QED}) \quad (1)$$

(ii)

Sketch:



$$A(x) = \frac{5}{2} (2x)^2 \frac{\sin^4 54^\circ}{\sin 72^\circ}$$

$$\therefore A(y) = 10y \frac{\sin^4 54^\circ}{\sin 72^\circ}$$

$$\therefore V = \lim_{\delta y \rightarrow 0} \sum \frac{b}{a} 10y \frac{\sin^4 54^\circ}{\sin 72^\circ} \delta y$$

$$= 10 \frac{\sin^4 54^\circ}{\sin 72^\circ} \int_0^3 y dy$$

$$= 5 \frac{\sin^4 54^\circ}{\sin 72^\circ} [y^2]_0^3$$

$$= 45 \frac{\sin^4 54^\circ}{\sin 72^\circ} \quad (= 30.968 \dots)$$

$$\therefore V = 31.0 \text{ u}^3 \quad (\text{Idp}) \quad (3)$$

$$c) (i) \delta V = \pi \{ (x+\delta x - 1)^2 - (x-1)^2 \} 2y$$

$$= 2\pi y \{ 2(x-1) + \delta x \} \delta x$$

$$= 4\pi (x-1) y \delta x \quad (\delta x^2 \text{ negligible})$$

$$= 4\pi (x-1) \sqrt{1-(x-2)^2} \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} 4\pi \sum \frac{b}{a} (x-1) \sqrt{1-(x-2)^2} \delta x$$

$$= 4\pi \int_1^3 (x-1) \sqrt{1-(x-2)^2} dx \quad (\text{QED}) \quad (3)$$

$$(ii) \text{ let } x-2 = \sin \theta \quad \therefore dx = \cos \theta d\theta$$

$$\text{at } x=1 \quad \sin \theta = -1 \quad \therefore \theta = -\pi/2$$

$$x=3 \quad \sin \theta = 1 \quad \therefore \theta = \pi/2$$

$$\therefore V = 4\pi \int_{-\pi/2}^{\pi/2} (1+\sin \theta) \sqrt{1-\sin^2 \theta} \cos \theta d\theta$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} (1+\sin \theta) \cos^2 \theta d\theta$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} \frac{1}{2} (1+\cos 2\theta) + \sin \theta \cos^2 \theta d\theta$$

$$= 4\pi \left[\frac{1}{2} (\theta + \frac{1}{2} \sin 2\theta) - \frac{1}{3} \cos^3 \theta \right]_{-\pi/2}^{\pi/2}$$

$$= 4\pi \left(\frac{\pi}{4} + 0 - 0 + \frac{\pi}{4} + 0 - 0 \right)$$

$$= 2\pi^2 \text{ u}^3$$

(3)

$$\textcircled{7} \text{ a) (i) } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore \frac{2x}{a^2} - \frac{2y}{b^2} y' = 0$$

$$\begin{aligned} \therefore y' &= \frac{b^2 x}{a^2 y} \\ &= \frac{b^2 \times a \sec \theta}{a^2 \times b \tan \theta} \\ &= \frac{b \sec \theta}{a \tan \theta} \end{aligned}$$

$\therefore$ Equin of tangent:

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$\therefore a \tan \theta y - a b \tan^2 \theta = b \sec \theta x - a b \sec^2 \theta$$

$$\therefore b \sec \theta x - a \tan \theta y = ab (\sec^2 \theta - \tan^2 \theta)$$

$$\therefore \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} - 1 = 0 \quad (\text{QED}) \quad \textcircled{2}$$

(ii) Perpendicular dist. from $(ae, 0)$ to circle tangent = radius

$$\therefore \frac{|e \sec \theta + 0 - 1|}{\sqrt{\frac{\sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}}} = a \sqrt{e^2 + 1}$$

$$\begin{aligned} \text{LHS} &= \frac{|e \sec \theta - 1|}{\sqrt{\frac{\sec^2 \theta}{a^2} + \frac{\sec^2 \theta - 1}{b^2}}} \\ &= \frac{|e \sec \theta - 1|}{\sqrt{\frac{\sec^2 \theta (a^2 + b^2)}{a^2 b^2} - \frac{1}{b^2}}} \end{aligned}$$

$$\text{but } \frac{b^2}{a^2} + 1 = e^2 \therefore a^2 + b^2 = a^2 e^2$$

$$= \frac{|e \sec \theta - 1|}{\sqrt{\frac{\sec^2 \theta (a^2 e^2)}{a^2 b^2} - \frac{1}{b^2}}}$$

$$= \frac{|e \sec \theta - 1|}{\sqrt{\frac{e^2 \sec^2 \theta - 1}{b^2}}}$$

$$\therefore \text{LHS}^2 = \text{RHS}^2$$

$$\therefore a^2 (e^2 + 1) = \frac{b^2 (e \sec \theta - 1)^2}{e^2 \sec^2 \theta - 1}$$

$$\therefore \frac{a^2}{b^2} (e^2 + 1) = \frac{e \sec \theta - 1}{e \sec \theta + 1}$$

$$\text{but } \frac{a^2}{b^2} = \frac{1}{e^2 - 1}$$

$$\therefore \frac{e^2 + 1}{e^2 - 1} = \frac{e \sec \theta - 1}{e \sec \theta + 1}$$

$$\therefore e^3 \sec \theta + e \sec \theta + e^2 + 1 = e^3 \sec \theta - e \sec \theta - e^2 + 1$$

$$\therefore 2e \sec \theta = -2e^2$$

$$\therefore \sec \theta = -e \quad (\text{QED}) \quad \textcircled{3}$$

(iii) Coordinates of P and Q are $(a \sec \theta, b \tan \theta)$

$$\therefore \text{if } \sec \theta = -e \text{ then } \sec^2 \theta = e^2$$

$$\therefore \tan^2 \theta + 1 = e^2$$

$$\therefore \tan \theta = \pm \sqrt{e^2 - 1}$$

$$\therefore P, Q \text{ are } (-ae, \pm b \sqrt{e^2 - 1})$$

i.e. on latus rectum

$$(iv) \text{ Tangents: } \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$\therefore$ Common tangents:

$$-\frac{x e}{a} - \frac{y (\pm \sqrt{e^2 - 1})}{b} = 1$$

$$\therefore -x e - y \left(\frac{\pm a \sqrt{e^2 - 1}}{b} \right) = a$$

$$\text{but } 1 = \frac{a^2}{b^2} (e^2 - 1)$$

$$\therefore x e \pm y + a = 0$$

$$\text{or } y = \pm (x e + a) \quad \textcircled{4}$$

a) (iv) continued:

solve simultaneously with circle equation:

$$\text{i.e. with: } y = \pm \sqrt{a^2(e^2+1) - (x-ae)^2}$$

$$\therefore (x+ae)^2 = a^2(e^2+1) - (x-ae)^2$$

$$\therefore x^2e^2 + 2axe + a^2 = a^2e^2 + a^2 - x^2 - 2axe - a^2e^2$$

$$\therefore x^2(e^2-1) = 0$$

$$\therefore x = 0$$

$$\text{for } x=0 \quad y = \pm a$$

$$\therefore \text{contact pts: } (0, \pm a) \quad (1)$$

b) (i) Prove:

$$\frac{\sin(A+B) - \sin(A-B)}{2\sin B} = \cos A$$

$$\text{LHS} = \frac{\sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B}{2\sin B}$$

$$= \frac{2\cos A \sin B}{2\sin B}$$

$$= \cos A$$

$$= \text{RHS (QED)} \quad (2)$$

(ii) Letting $A = (2n-1)\pi$, $B = \pi$, then:

$$\cos(2n-1)\pi = \frac{\sin 2n\pi - \sin 2(n-1)\pi}{2\sin \pi}$$

$$= \frac{\sin 2n\pi}{2\sin \pi} - \frac{\sin 2(n-1)\pi}{2\sin \pi}$$

$$\therefore \cos \pi + \cos 3\pi + \cos 5\pi + \dots + \cos(2n-3)\pi + \cos(2n-1)\pi$$

$$= \left(\frac{\sin 2\pi}{2\sin \pi} - \frac{\sin 0}{2\sin \pi} \right) + \left(\frac{\sin 4\pi}{2\sin \pi} - \frac{\sin 2\pi}{2\sin \pi} \right) + \left(\frac{\sin 6\pi}{2\sin \pi} - \frac{\sin 4\pi}{2\sin \pi} \right) + \dots + \left(\frac{\sin 2(n-1)\pi}{2\sin \pi} - \frac{\sin 2(n-2)\pi}{2\sin \pi} \right) + \left(\frac{\sin 2n\pi}{2\sin \pi} - \frac{\sin 2(n-1)\pi}{2\sin \pi} \right)$$

$$= \frac{\sin 2n\pi}{2\sin \pi} \quad (\text{QED}) \quad (3)$$

$$(iii) \int_0^{\frac{\pi}{2}} \frac{\sin 8x}{\sin x} dx$$

$$= 2 \int_0^{\frac{\pi}{2}} \frac{\sin 2 \times 4x}{2\sin x} dx \quad (\text{i.e. } n=4)$$

$$= 2 \int_0^{\frac{\pi}{2}} (\cos x + \cos 3x + \cos 5x + \cos 7x) dx$$

(from (ii))

$$= 2 \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \frac{1}{7} \sin 7x \right]_0^{\frac{\pi}{2}}$$

$$= 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} \right)$$

$$= \frac{152}{105} \quad (2)$$

⑧ a) (i) In $\triangle ADB$: letting $BD = x$:

$$\frac{x}{\sin \theta} = \frac{AB}{\sin ADB}$$

$$\text{In } \triangle ADC: DC = 1-x$$

$$\frac{1-x}{\sin 2\theta} = \frac{AC}{\sin ADC}$$

$$\text{but } \sin ADC = \sin(180^\circ - \angle ADB) = \sin ADB$$

$$\text{and } AB = AC$$

$$\therefore \frac{AC}{\sin ADC} = \frac{AB}{\sin ADB}$$

$$\therefore \frac{x}{\sin \theta} = \frac{1-x}{\sin 2\theta}$$

$$\therefore \frac{2\sin \theta \cos \theta}{\sin \theta} = \frac{1-x}{x}$$

$$\therefore \cos \theta = \frac{1-x}{2x} \quad (\text{QED}) \quad (4)$$

(ii) $0^\circ < 3\theta < 180^\circ$

$$\therefore 0^\circ < \theta < 60^\circ$$

$$\therefore 1 > \cos \theta > \frac{1}{2}$$

$$\therefore 1 > \frac{1-x}{2x} > \frac{1}{2} \quad (x > 0)$$

$$2x > 1-x > x$$

$$3x > 1 > 2x$$

$$x > \frac{1}{3} : x < \frac{1}{2} \Rightarrow \therefore \frac{1}{3} < x < \frac{1}{2} \quad (\text{QED}) \quad (5)$$

⑧ b) (i) $\alpha + \beta + \gamma = 0$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 3p$$

$$\alpha\beta\gamma = -q$$

$$\frac{\alpha\beta}{\gamma} + \frac{\alpha\gamma}{\beta} + \frac{\beta\gamma}{\alpha} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma}$$

$$= \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma}$$

$$= -\frac{9p^2}{q}$$

AND:

$$\frac{\alpha\beta}{\gamma} \cdot \frac{\alpha\gamma}{\beta} + \frac{\alpha\beta}{\gamma} \cdot \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} \cdot \frac{\beta\gamma}{\alpha} = \alpha^2 + \beta^2 + \gamma^2$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= -6p$$

AND

$$\frac{\alpha\beta}{\gamma} \times \frac{\alpha\gamma}{\beta} \times \frac{\beta\gamma}{\alpha} = \alpha\beta\gamma$$

$$= -q$$

$\therefore$ equation is:

$$x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0$$

③

(ii) If $\alpha\beta = \gamma$ then the root: $\frac{\alpha\beta}{\gamma}$ is 1

$$\therefore 1 + \frac{9p^2}{q} - 6p + q = 0$$

$$\therefore q + 9p^2 - 6pq + q^2 = 0$$

$$\therefore (3p - q)^2 + q = 0 \quad (\text{QED}) \quad ②$$

c) (i) $\cos x = \cos(-x)$

$$= \sin\left(\frac{\pi}{2} - (-x)\right)$$

$$= \sin\left(\frac{\pi}{2} + x\right) \quad (\text{QED})$$

$$\{ \text{OR} \} \sin\left(x + \frac{\pi}{2}\right) = \sin x \cos \frac{\pi}{2} + \cos x \sin \frac{\pi}{2}$$

$$= \cos x \}$$

For $n=1$: Prove -

$$\frac{d}{dx} (5 \sin(x + \alpha)) = 5 \sin\left(x + \alpha + \frac{\pi}{2}\right)$$

$$\text{LHS} = 5 \cos(x + \alpha)$$

$$\text{RHS} = 5 \sin\left(x + \alpha + \frac{\pi}{2}\right) \quad \left\{ \text{from (i)} \right.$$

$$= 5 \cos(x + \alpha)$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{i.e. } \frac{d^k y}{dx^k} = 5 \sin\left\{ (x + \alpha) + \frac{k\pi}{2} \right\}$$

Now prove true for $n=k+1$

i.e. prove:

$$\frac{d^{k+1} y}{dx^{k+1}} = 5 \sin\left\{ (x + \alpha) + (k+1)\frac{\pi}{2} \right\}$$

$$\text{LHS} = \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right)$$

$$= \frac{d}{dx} \left(5 \sin\left\{ (x + \alpha) + \frac{k\pi}{2} \right\} \right) \quad \left\{ \text{from assumption} \right.$$

$$= 5 \cos\left\{ (x + \alpha) + \frac{k\pi}{2} \right\}$$

$$= 5 \sin\left\{ (x + \alpha) + \frac{k\pi}{2} + \frac{\pi}{2} \right\} \quad \left\{ \text{from (i)} \right.$$

$$= 5 \sin\left\{ (x + \alpha) + (k+1)\frac{\pi}{2} \right\}$$

$$= \text{RHS}$$

$\therefore$ True for $n=1$ and true for $n=k+1$
when true for $n=k$

$\therefore$ True for all $n \geq 1$ (QED)

④